TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Miniaturized bioprocess tools for high-throughput perfusion-like cell culture**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio-Medical		Bio-Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-equiment	Bio-process technologies Bio-process equipment technologies	Medical-Bio	Pharmaceutical bio-based technologies and systems
Concept of Proposed Technology	We propose a platform of miniaturized cell-retention bioprocess tools enabling semi-continuous perfusion-like operation in standard microbioreactor formats [e.g., 24-well deep well plates (DWPs), 50 mL tubes] with high sample throughput and low working volumes. The platform covers: 1. Cell-retention pipette-tip enabling one-step medium exchange with ≥95 % cell retention and gentle handling to preserve viability, compatible with multi-channel electronic pipettes and/or common liquid-handling robots. 2. DWP insert with integrated micro-membrane and automated parallelized fluid handling unit, to mimic tangential-flow filtration (TFF) at well scale without the necessity to interrupt the orbital agitation Both approaches allow repeated partial harvest and feed in DWPs without centrifugation, and aim to bridge fed-batch screens to perfusion-like conditions while keeping costs and complexity low. Potential uses: - Upstream process development such as antibody or adeno-associated virus production and other high-density cultures - DoE screening of perfusion parameters - Scale-down models aligned with continuous/perfusion production strategies			

Rationale for Joint R&D Project	The bioprocessing industry is undergoing a major transition from traditional fed-batch operation toward perfusion and semi-continuous perfusion, driven by the need for higher productivities, improved product quality, and more stable long-term culture conditions. However, despite this strategic shift, high-throughput and low-volume process-development tools remain fundamentally limited: existing microbioreactor formats either lack integrated cell-retention capabilities or offer only low sample throughput at larger volumes, preventing efficient exploration of the wide parameter space required for perfusion optimization. As documented in current market and technology assessments (1), the absence of robust miniaturized cell-retention solutions in standard labware (e.g., DWPs, small tubes, robotic consumables) forces developers to rely on cumbersome centrifugation or stop-settle workflows, which induce cell stress, reduce viability, increase contamination risk, and restrict medium-exchange frequency to as low as one to four exchanges per day, even with automation. This creates a severe bottleneck for parameter screening, as true perfusion conditions require much more frequent medium refresh and gentle handling. Consequently, scaled-down models at microscale often fail to accurately reproduce the biochemical and hydrodynamic profiles of larger perfusion bioreactors, thereby compromising scale-down fidelity, slowing development cycles, and increasing overall cost and risk. A joint R&D effort between CSEM and Korean SMEs would directly address this recognized gap by co-developing high-throughput, miniaturized cell-retention technologies that enable perfusion-like operation in widely adopted labware formats, supporting the global trend toward continuous bioprocessing.
Key Words	1. Miniaturized bioreactors 2. Cell retention 3. Perfusion screening 4. Membrane-integrated labware 5. High-throughput upstream processing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3-4	End of the Project	Level 5-6
R&D Objectives	**1.** Design & prototype a modular miniaturized bioprocess platform: • A. Pipette-tip cell-retention consumable (semi-continuous medium exchange) for semi-continuous medium exchange under short-time static conditions • B. DWP TFF-like insert & fluid handling unit enabling on-plate retention/harvest workflows under agitation. **2.** Membrane selection & integration for 3D geometries (cones/cylinders/inserts) compatible with scalable sealing/assembly process, preserving porosity and biocompatibility. **3.** Define operating protocols for multi-channel electronic pipettes and benchtop liquid handlers (flow rates, immersion profiles, wash steps) enabling ≥4–8 exchanges/day at well scale without centrifugation. **4.** Quantified performance in representative use cases: ≥95 % cell retention, viability ≥95 %, volume CV ≤10 %, effective perfusion rates supporting high cell density operation. **5.** Scalability & manufacturability: design for injection molding, minimum parts/assembly, sterilization compatibility (autoclave, EtOH dip, e-beam/gamma, EtO), and target cost ≈ 1 CHF per tip/insert at scale.			
R&D Target Specifications	• Cell retention: ≥95 % across 40-60×10^6 cells/mL; pressure drop ≤0.5 bar across membrane • Membranes : 0.2-7 µm pore-size options; ≥3 cycles clog-resistance at 60×10^6 cells/mL • Volume accuracy: aspiration/dispense ±10 %; per DWP row/column <2 min operation • Sterilization robustness: at least 1 validated method (e.g., autoclave or EtOH) • Manufacturing: injection-moldable bodies (PP/PS/PC/COC), membrane bonding validated on curved surfaces; minimum parts per assembly.			
R&D Plan	**WP1 – Membrane scouting & integration strategy (M1–M6)** • Bench tests for cell retention, burst pressure, fouling, sterilization resilience on candidate membranes (PES, cellulose, PC, nylon, PTFE-based) with flow-through jigs. • Bonding trials with various upscale methods, define design rules for tip and DWP-insert geometries **WP2 – Prototyping of two lines (M7–M14)** • **Line A: Pipette-tip with cell-retention**: geometry tuning for electronic pipette and liquid handling robot compatibility, membrane positioning & sealing; sterilization tests; operating SOP (flow/immersion/wash). • **Line B: DWP TFF-like insert**: insert with adapted geometry and side-wall membrane windows to drive cross-flow-like exchange under continuous perfusion; dead-volume minimization; liquid handling system including DWP-insert interface for continuous perfusion • Deliverables: Sterile prototypes (≥24 units/line) + functional report (volume CV, retention, robustness). **WP3 – Application validation (M14–M18)**			

	• Relevant use case applied to small-scale runs in standard labs (DWP / 50 mL tubes), mirrored comparisons vs. pseudo-perfusion (centrifugation and medium exchange by automated pipetting). • KPIs: ≥95 % retention & viability, reduced handling time/steps, contamination incidence, product qualification and DoE throughput gains. • Deliverables: PoC dossiers for each use case; industrialization brief (Line A & B). **Milestones** • **M6 Go/No-Go:** ≥1 membrane + ≥1 sealing method validated under wet conditions; preliminary sterilization compatibility; design rules frozen. • **M14 Prototype readiness:** prototypes meet ±10 % volume reproducibility within one run and across runs, ≥95 % cell retention in at least one cycle post-sterilization; SOPs finalized; ready for application validation.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• System concept & architecture for the miniaturized bioprocess platform (pipette-tip + DWP insert). • Membrane integration R&D on 3D geometries; bonding tool design and test jigs; sterilization studies. • Prototype design & fabrication (additive, machining, pilot molding) and SOPs for medium exchange with multi-channel e-pipettes / benchtop handlers. • Functional validation (volume metrology, retention bench tests with fixed cells), data analysis, IP landscaping, and industrialization guidelines.
Expected R&R of Korean SME Partner	Korean SME partners may come from multiple complementary profiles, enabling a robust industrialization and validation pathway: **1. Precision plastics / membrane-integration manufacturers** • Adapt CSEM concept designs (pipette-tip and DWP insert) for injection-mold tooling, multi-cavity molds, and scalable membrane-integration fixtures. • Develop membrane-sealing process optimized for the 3D geometries used in the miniaturized bioprocess tools. • Validate sterilization-ready production workflows (gamma, EtO, e-beam) and consumable packaging. **2. Liquid-handling automation SMEs** • Ensure hardware/consumable compatibility (pipette arm tolerances, force limits, deck calibration). • Co-develop automation scripts/macros enabling semi-continuous perfusion cycles in standard and high-throughput formats. • Validate execution of ≥4-8 medium-exchange cycles/day in fully automated workflows. **3. Microbioreactor platform & consumables vendors** • Provide interface specifications so that the DWP TFF-like insert integrates mechanically and fluidically into existing microbioreactor ecosystems. • Perform co-validation of perfusion-like operation under platform-specific agitation, environmental control, and sampling conditions. • Explore bundled commercialization (e.g., “next-gen perfusion screening kit” combining inserts + platform workflow). **4. Biological-validation SMEs (bioprocess CDMOs, assay houses, HT biolabs)** • Conduct independent validation of cell retention, viability, and product yield (using e.g. CHO culture, gold standard for mAb production) under realistic process conditions. • Generate application notes, performance fingerprints, and comparative data versus pseudo-perfusion workflows. • Support early industry adoption with testimonial data and real-world use cases. **Cross-cutting SME roles across all profiles:**

	• Participate in joint IP development, share manufacturability insights, and co-file patents where appropriate. • Contribute to cost-down analysis targeting ~1 CHF per disposable unit at volume. • Support commercialization in Korea and the broader Asian bioprocessing market. • Provide market intelligence on regional adoption drivers, including cell therapy, gene therapy, and antibody biomanufacturing sectors.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements** • Two validated product lines (pipette-tip retention; DWP TFF-like insert) at TRL5–6, with SOPs, design rules, and industrialization briefs. • Quantified gains: ≥33 % shorter optimization time, significant OpEx reduction from low-volume, high-throughput screens, and improved scale-down fidelity to perfusion conditions. **Commercialization Potential** • Consumable business model with recurring revenue; compatibility with existing benchtop automation enables rapid adoption; dual route: universal consumables and platform-specific integration kits. **Broader Industrial/Economic Impact** • Swiss - Korean value chain for advanced bioprocess labware; local manufacturing & skilled jobs; export-ready products. • Sustainability: reduced media/consumables via low-volume workflows; fewer centrifugation steps, i.e. lower contamination risk and energy use. • Societal: faster, more consistent development of biological and advanced therapies, potentially improving accessibility.

6. References

(1) Romann, P., Giller, P., Sibilia, A., Herwig, C., Zydney, A. L., Perilleux, A., Souquet, J., Bielser, J.-M., & Villiger, T. K. (2023). Co-current filtrate flow in TFF perfusion processes: Decoupling transmembrane pressure from crossflow to improve product sieving. Biotechnology & Bioengineering, 121(2), 640–654. https://doi.org/10.1002/bit.28589

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization